

Rojae Mighty

rojae.mighty@stonybrook.edu · Stony Brook, NY · rojaemighty.com · linkedin.com/in/rojaemight

Education

Stony Brook University

B.S. in Physics (Natural Sciences)

Expected Graduation: 5/28

Programming languages: **Python**, **C/C++** · Relevant coursework: statistical mechanics, computational methods, data analysis, probability

Research Experience

Brookhaven National Laboratory

2025–Present

- **Department of Energy Science Undergraduate Laboratory Internship (SULI)** (Summers 2025, 2026): Electron-ion collision studies of semi-inclusive processes; implemented a spatially dependent parton distribution function using **ROOT**, **Python**, and **Unix/Linux shell scripting**.

Center for Frontiers in Nuclear Science (CFNS)

2025–Present

Undergraduate Researcher · Advisor: Zhoudunming (Kong) Tu

- Developed **machine learning** pipelines in **Python** with **scikit-learn** for kinematic reconstruction on large-scale collision datasets; applied **data science** methods including **feature engineering** and **statistical data analysis**.
- Used **C++** and **ROOT** with **Monte Carlo event generators** to study gluon saturation through geometric scaling; ran **numerical simulation** studies of high-energy nuclear collision phenomena.
- Delivered **conference presentations** at DIS 2026, APS Global Physics Summit (oral), and Stony Brook Physics Colloquium; prepared materials in **LaTeX**.

Projects

Market Microstructure Forecasting

2025–Present · *GitHub*

- Built a **Python machine learning** pipeline with **pandas**, **NumPy**, and **scikit-learn**; **feature engineering** and **walk-forward validation** on time-series limit order book data; **Git**-versioned on GitHub.

J.P. Morgan Quantitative Research (Forage)

2026

- Forage certificate: commodity **statistical data analysis**; **derivatives pricing**; **Python** credit-risk model; FICO bucketing via **dynamic programming/scikit-learn** (MSE/log-likelihood).

Honors & Presentations

- 1st Place, Physical Sciences Presentation, Yale Undergraduate Research Conference (2026)
- Outstanding Oral Presentation Award, National Society of Black Physicists Conference (2025)
- Barry Barish Undergraduate Research Travel Award, Stony Brook University (2026)
- *Geometric Scaling and Gluon Saturation at the EIC* — DIS 2026, Yale URC, NSBP, APS DNP (CEU), APS Global Physics Summit, Stony Brook Colloquium

Technical Skills

Languages: Python, C/C++, Unix/Linux shell scripting

Libraries: pandas, NumPy, scikit-learn

ML & data: machine learning, data science, statistical data analysis, feature engineering, walk-forward validation, numerical simulation, dynamic programming

Tools: ROOT, Git, LaTeX, Monte Carlo event generators